LLM pricing verification for ContextKeeper WP-003

The model pricing landscape has shifted dramatically since mid-2025. Several models referenced in the whitepaper are now deprecated or superseded, flagship models have changed across all providers, and a new generation of 1M+ context window models has become the standard. This verification identifies **27 specific corrections** needed for WP-003, with exact pricing confirmed from official sources as of March 18, 2026.

The most critical finding: **Llama 3.1 70B is no longer available on Groq, Together AI, or Fireworks AI**—replaced by Llama 3.3 70B at higher prices. Claude 3.5 Haiku, Claude 3.7 Sonnet, and Claude 3 Opus are all deprecated on Anthropic's API. claude OpenAI's flagship is now GPT-5.4, Aicritique and Google has progressed to Gemini 3.1 Pro Preview. Google AI google The whitepaper's illustrative cost table requires complete recalculation.

Priority 1: OpenAI pricing and model specifications

GPT-4o (still available in API, retired from ChatGPT Feb 13, 2026)

Metric	Value	Source
Input	**$2.50/1M tokens**	developers.openai.com/api/docs/pricing
Cached input	$1.25/1M tokens	Same
Output	**$10.00/1M tokens**	Same
Context window	**128,000 tokens**	Unchanged
Max output	16,384 tokens	Same

Pricing has **not changed** since mid-2025 for the current `gpt-4o` alias. However, the original `gpt-4o-2024-05-13` snapshot remains available at a higher rate ($5.00/$15.00). openai The whitepaper should specify which snapshot it references.

GPT-4o mini

Metric	Value
Input	**$0.15/1M tokens**
Cached input	$0.075/1M

Metric	Value
Output	**$0.60/1M tokens**
Context window	**128,000 tokens**
Max output	16,384 tokens

GPT-4.1 family (released April 14, 2025)

OpenAI released **GPT-4.1, GPT-4.1 mini, and GPT-4.1 nano** with a breakthrough **1,047,576-token context window (~1.05M)** and 32,768 max output. OpenRouter These models remain available in the API despite being retired from ChatGPT on February 13, 2026. OpenAI

Model	Input/1M	Cached input/1M	Output/1M	Context
GPT-4.1	**$2.00**	$0.50	**$8.00**	1.05M
GPT-4.1 mini	**$0.40**	$0.10	**$1.60**	1.05M
GPT-4.1 nano	**$0.10**	$0.025	**$0.40**	1.05M

GPT-4.1 is **26% cheaper** than GPT-4o per token OpenAI with an 8× larger context window— kaelresearch a critical update for the whitepaper's cost modeling.

Reasoning models (o-series)

Model	Input/1M	Cached input/1M	Output/1M	Context	Max output
o3 (Apr 16, 2025)	**$2.00**	$0.50	**$8.00**	200K	100K
o4-mini (Apr 16, 2025)	**$1.10**	$0.275	**$4.40**	200K	100K
o3-mini	**$1.10**	$0.55	**$4.40**	200K	100K
o1	**$15.00**	$7.50	**$60.00**	200K	100K
o3-pro (Jun 10, 2025)	**$20.00**	—	**$80.00**	200K	—

Reasoning token billing: Reasoning tokens are **invisible in the API** but occupy context window space and are **billed as output tokens.** Simon Willison The breakdown appears in `completion_tokens_details.reasoning_tokens` (Chat Completions) or `output_tokens_details.reasoning_tokens` (Responses API). Openai

Current OpenAI flagship: GPT-5.4 (released March 5, 2026)

GPT-5.4 is OpenAI's most capable model OpenAI with **tiered context pricing:** Silicondata

Model	Input/1M (≤272K)	Input/1M (>272K)	Output/1M (≤272K)	Output/1M (>272K)	Context
GPT-5.4	**$2.50**	$5.00	**$15.00**	$22.50	1.05M
GPT-5.4 pro	$30.00	$60.00	$180.00	$270.00	1.05M

GPT-5.4 mini ($0.25/$2.00, 400K context) and GPT-5.4 nano (~$0.20/$1.25) were released **March 17, 2026**—one day before this research. Official API pricing for mini/nano may still be propagating.

Most cost-effective OpenAI model for high-volume agent runs

GPT-5-nano at **$0.05 input / $0.40 output per 1M tokens** is the cheapest current-generation model. kaelresearch Curlscape For tasks requiring a 1M context window, **GPT-4.1 nano** ($0.10/$0.40) offers the best value. For reasoning-intensive agent workflows, **o4-mini** ($1.10/$4.40) is the most cost-effective reasoning model. kaelresearch Kael Research Batch API provides an additional 50% discount across all models. Curlscape

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Priority 1: Anthropic pricing and model specifications

Critical finding: most whitepaper models are deprecated

Claude 3.5 Haiku, Claude 3.7 Sonnet, Claude 3.5 Sonnet, and Claude 3 Opus are all retired on Anthropic's API as of March 2026. Requests to these models return errors. The whitepaper must be updated to reference the Claude 4 family.

Claude 3.5 Haiku (DEPRECATED — retired, requests return errors)

Metric	Value	Status
Input	$0.80/1M tokens	Deprecated
Output	$4.00/1M tokens	Deprecated
Context window	**200,000 tokens** ✅	Whitepaper claim CONFIRMED

Metric	Value	Status
Max output	8,192 tokens	—

On Vertex AI, deprecated January 5, 2026 with shutdown July 5, 2026. Google

Claude 3.7 Sonnet (DEPRECATED — retired)

Metric	Value
Input	$3.00/1M tokens
Output	$15.00/1M tokens
Context window	**200,000 tokens**
Max output	64,000 tokens (128K with extended thinking)
Release date	**February 24, 2025**

First hybrid reasoning model from Anthropic—introduced extended thinking mode. On Vertex AI, deprecated November 11, 2025, shutdown May 11, 2026. Google Replacement: Claude Sonnet 4.6.

Claude 3.5 Sonnet — fully retired, no longer available

Both versions (`claude-3-5-sonnet-20240620` and `claude-3-5-sonnet-20241022`) return errors. On Vertex AI, deprecated August 20, 2025, shut down February 19, 2026. Superseded by Claude Sonnet 4/4.5/4.6.

Claude 3 Opus — deprecated, no longer flagship

Priced at $15.00/$75.00 per 1M tokens. nOps claude Deprecated January 2026 on Anthropic API; briefly brought back for paying customers by request. Superseded by Claude Opus 4, 4.1, 4.5, and 4.6.

Current Anthropic flagship: Claude Opus 4.6 (released February 5, 2026)

Model	Input/1M	Output/1M	Context	Max output	Status
Claude Opus 4.6	**$5.00**	**$25.00**	**1M (GA)**	128K	**Flagship**
Claude Opus 4.5	$5.00	$25.00	200K (1M beta)	128K	Active
Claude Opus 4/4.1	$15.00	$75.00	200K	—	Active (legacy)

Model	Input/1M	Output/1M	Context	Max output	Status
Claude Sonnet 4.6	**$3.00**	**$15.00**	**1M (GA)**	128K	Active
Claude Sonnet 4.5	$3.00	$15.00	200K (1M beta)	—	Active
Claude Sonnet 4	$3.00	$15.00	200K (1M beta)	—	Active
Claude Haiku 4.5	**$1.00**	**$5.00**	200K	64K	Fastest/cheapest
Claude Haiku 3	$0.25	$1.25	200K	4,096	Retiring Apr 19, 2026

Opus pricing dropped 67%: From $15/$75 (Opus 3/4/4.1) to **$5/$25** (Opus 4.5/4.6). Silicondata IntuitionLabs Sonnet pricing has been **remarkably stable at $3/$15** across six versions. IntuitionLabs As of ~March 14, 2026, Anthropic removed the long-context pricing surcharge for Opus 4.6 and Sonnet 4.6, making **1M tokens available at standard per-token rates** without a beta header. The New Stack

Extended thinking pricing — confirmed

Extended thinking tokens are billed as output tokens at standard rates. There is no separate pricing tier. MetaCTO For Claude Opus 4.6, "adaptive thinking" (`thinking: {type: "adaptive"}`) replaces manual budget setting—the model dynamically decides when deep reasoning is needed, controlled by an `effort` parameter (low/medium/high/max). You are charged for the **full thinking tokens generated**, not just the summary tokens returned.

Priority 1: Google Gemini pricing and model specifications

Gemini 2.0 Flash (DEPRECATED — shutdown June 1, 2026)

Metric	Value
Input	**$0.10/1M tokens**
Output	**$0.40/1M tokens**
Context window	**1,048,576 tokens (1M)** ✅
Max output	8,192 tokens

Still functional but scheduled for shutdown June 1, 2026. google Firebase

Gemini 2.0 Pro — never reached GA

Released only as an experimental model (`gemini-2.0-pro-exp-02-05`) on February 5, 2025. Google AI Wikipedia No production pricing was established. Superseded by Gemini 2.5 Pro (GA June 17, 2025). Wikipedia

Gemini 2.5 Pro (GA — current stable flagship)

Metric	≤200K context	>200K context
Input	**$1.25/1M**	$2.50/1M
Output (incl. thinking)	**$10.00/1M**	$15.00/1M
Context window	**1,048,576 (1M)**	Same
Max output	65,536 tokens	Same

This is a **native thinking model**—chain-of-thought reasoning is built in, Google controllable via `thinkingBudget` parameter. Thinking tokens are billed at the output rate.

Gemini 2.5 Flash (GA — best value hybrid reasoning)

Metric	Value
Input	**$0.30/1M tokens**
Output (incl. thinking)	**$2.50/1M tokens**
Context window	**1,048,576 (1M)**
Max output	65,535 tokens

Gemini 3.1 Pro Preview (latest, released ~March 2026)

Metric	≤200K context	>200K context
Input	**$2.00/1M**	$4.00/1M
Output (incl. thinking)	**$12.00/1M**	$18.00/1M
Context window	1,048,576 (1M)	Same

Still in preview status. IntuitionLabs No free tier. google Supports thinking levels: 'low', 'medium', 'high'. kaelresearch

Most cost-effective Gemini for enterprise

Gemini 2.5 Flash-Lite at **$0.10 input / $0.40 output per 1M tokens** is the most cost-effective stable model google with 1M context. For reasoning tasks, **Gemini 2.5 Flash** at $0.30/$2.50 provides the best price-performance with built-in thinking. Costgoat Google's **Vertex AI Model Optimizer** can dynamically route requests between models to optimize cost vs. quality. Google Cloud

Gemini 1.5 Pro is fully retired

All Gemini 1.5 models shut down by September 24, 2025. All requests return 404 errors.

Priority 1: Open-source and alternative inference pricing

Meta Llama — current status

Llama 3.1 is not the latest. The current Llama family includes:

- **Llama 3.2** (September 2024): 1B, 3B, 11B, 90B. Lightweight + multimodal. 128K context. AWS
- **Llama 3.3** (December 2024): **70B instruct model**, 128K context. Performance comparable to Llama 3.1 405B. DataCamp
- **Llama 4 Scout** (April 5, 2025): 17B active × 16 experts (MoE). 128K standard context (10M claimed). Natively multimodal. CodingScape
- **Llama 4 Maverick** (April 5, 2025): 17B active × 128 experts (~401B total). 128K standard (1M claimed). Natively multimodal. CodingScape
- **Llama 4 Behemoth**: Still in training/ TechCrunch unreleased as of March 2026.

Groq pricing (confirmed March 18, 2026 from groq.com/pricing)

Model	Input/1M	Output/1M	Speed (tok/s)
Llama 4 Scout	**$0.11**	**$0.34**	594
Llama 4 Maverick	**$0.20**	**$0.60**	562
Llama 3.3 70B Versatile	**$0.59**	**$0.79**	394

Model	Input/1M	Output/1M	Speed (tok/s)
Llama 3.1 8B Instant	$0.05	$0.08	840
Qwen3 32B	$0.29	$0.59	662

⚠️ **Critical whitepaper correction: Llama 3.1 70B is NOT listed on Groq's current pricing page.** The 70B-class model is now **Llama 3.3 70B Versatile** at $0.59/$0.79—significantly more expensive than the whitepaper's assumed pricing. Only Llama 3.1 8B remains. groq

Together AI pricing (confirmed March 18, 2026)

Model	Input/1M	Output/1M
Llama 4 Maverick	**$0.27**	**$0.85**
Llama 3.3 70B	**$0.88**	**$0.88**
Llama 3 8B Instruct Lite	$0.10	$0.10
DeepSeek R1-0528	$3.00	$7.00
DeepSeek V3.1	$0.60	$1.70

Fireworks AI pricing (confirmed March 18, 2026)

Fireworks uses tiered pricing: Llama 3.3 70B (>16B parameters, non-MoE) falls under the **$0.90/1M blended** tier. Llama 4 Scout/Maverick (MoE 0–56B active) falls under $0.50/1M blended. fireworks

DeepSeek — now a major player

DeepSeek V3.2 (released December 1, 2025) offers unified chat + reasoning at **$0.28 input / $0.42 output per 1M tokens** deepseek with 90% cache-hit discounts. This is competitive with open-source inference providers and should be included in the whitepaper's pricing table.

Priority 2: Context window verification

Model	Whitepaper claim	Verified value	Status
GPT-4o	128,000	**128,000** ✅	Unchanged

Model	Whitepaper claim	Verified value	Status
Claude 3.5 Haiku	200,000	**200,000** ✅	Confirmed correct
Gemini 2.0 Flash	1,000,000	**1,048,576** ✅	Confirmed (exactly 1M)
Llama 3.1 70B	131,072	**131,072** ✅	Confirmed
Claude 3.7 Sonnet	(not claimed)	**200,000**	All Claude 3.x share 200K

All context window claims in the whitepaper are confirmed accurate. However, the whitepaper should note that newer models have significantly expanded: GPT-4.1/5.4 now offer **~1.05M**, (TechCrunch) (OpenRouter) Claude Opus/Sonnet 4.6 offer **1M (GA)**, (ScreenApp) and all Gemini 2.5+ models offer **1M**. (Google)

Priority 3: Cost illustration recalculation

Using an accumulating context model (10 exchanges, each adding ~2,000 input tokens + 500 output tokens, with full conversation history resent each turn): **~132,500 total input tokens, 5,000 total output tokens across 10 exchanges.**

Updated cost table for the whitepaper

Model	Input rate/1M	Output rate/1M	Est. cost for 10 exchanges	Whitepaper original
GPT-4o	$2.50	$10.00	**~$0.38**	$0.47
GPT-4.1	$2.00	$8.00	**~$0.31**	N/A (new)
GPT-4.1 nano	$0.10	$0.40	**~$0.02**	N/A (new)
Claude Haiku 4.5	$1.00	$5.00	**~$0.16**	N/A (replacement for 3.5 Haiku)
Claude 3.5 Haiku (deprecated)	$0.80	$4.00	**~$0.13**	$0.08
Claude Sonnet 4.6	$3.00	$15.00	**~$0.47**	N/A (new)
Llama 3.3 70B on Groq	$0.59	$0.79	**~$0.08**	$0.03 (for 3.1 70B)
Llama 4 Scout on Groq	$0.11	$0.34	**~$0.02**	N/A (new)

Model	Input rate/1M	Output rate/1M	Est. cost for 10 exchanges	Whitepaper original
Gemini 2.5 Flash	$0.30	$2.50	~**$0.05**	N/A (new)
DeepSeek V3.2	$0.28	$0.42	~**$0.04**	N/A (new)

The whitepaper's original $0.47 for GPT-4o is **slightly high** at current pricing—$0.38 is more accurate with the $2.50/$10.00 rate. The original may have used the older `gpt-4o-2024-05-13` pricing ($5.00/$15.00). The $0.08 for Claude Haiku appears to have been based on **Claude 3 Haiku** ($0.25/$1.25), not Claude 3.5 Haiku. Costgoat The $0.03 for Llama 3.1 70B on Groq reflects historical pricing that no longer exists—the equivalent Llama 3.3 70B now costs ~**$0.08**. groq

Recommended replacement illustration for the whitepaper:

- GPT-4.1: ~$0.31 (mainstream quality, 1M context)
- Gemini 2.5 Flash: ~$0.05 (best value with reasoning)
- Llama 4 Scout on Groq: ~$0.02 (fastest, open-source)

Priority 4: Model landscape changes since mid-2025

Major new models launched

- **GPT-4.1** / **4.1 mini** / **4.1 nano** (April 14, 2025) — 1M context, cheaper than GPT-4o OpenAI
- **o3** / **o4-mini** Kael Research (April 16, 2025) OpenAI — replaced o1-preview and o1-mini Simon Willison
- **o3-pro** (June 10, 2025) — highest-cost reasoning model kaelresearch OpenAI
- **GPT-5** / **5-mini** / **5-nano** (2025) — next-generation family
- **GPT-5.4** / **5.4 pro** / **5.4 mini** / **5.4 nano** (March 5–17, 2026) — current flagship Fortune
- **Claude 3.7 Sonnet** (February 24, 2025) — first hybrid reasoning Claude
- **Claude Sonnet 4** / **Opus 4** through **Sonnet 4.6** / **Opus 4.6** (2025–February 2026)
- **Claude Haiku 4.5** — current fastest/cheapest Anthropic model IntuitionLabs +2
- **Gemini 2.5 Pro** (GA June 17, 2025) Wikipedia and **Gemini 2.5 Flash** — native thinking models
- **Gemini 3.1 Pro Preview** (~March 2026) — latest Google flagship kaelresearch +2
- **Llama 4 Scout** / **Maverick** (April 5, 2025) — MoE architecture, multimodal CodingScape
- **DeepSeek V3.2** (December 1, 2025) — unified chat+reasoning at $0.28/$0.42

- **OpenAI GPT-OSS-120B / GPT-OSS-20B** — open-weight models from OpenAI, a significant shift

Deprecated or no longer recommended

- **GPT-4.5 preview** — deprecated April 14, 2025, Portkey shut down July 14, 2025
- **o1-preview** — deprecated April 28, 2025, replaced by o3 Portkey
- **o1-mini** — deprecated April 28, 2025, replaced by o4-mini
- **Claude 3.5 Sonnet (both versions)** — fully retired, Google returns errors
- **Claude 3.5 Haiku** — retired, returns errors Claude API Docs
- **Claude 3.7 Sonnet** — retired, returns errors Claude API Docs
- **Claude 3 Opus** — deprecated, legacy access only
- **Gemini 1.5 Pro** — fully shut down (September 24, 2025), returns 404
- **Gemini 2.0 Flash** — deprecated, shutting down June 1, 2026 google Firebase
- **Llama 3.1 70B on hosted APIs** — replaced by Llama 3.3 70B on Groq, Together, Fireworks

Reasoning/thinking token billing across providers

All three major providers bill thinking tokens as output tokens at standard output rates. The key difference is transparency:

- **OpenAI**: Reasoning tokens hidden but counted in `completion_tokens`; breakdown in `completion_tokens_details.reasoning_tokens` OpenAI
- **Anthropic**: Thinking content visible as `type: "thinking"` content blocks; billed via `output_tokens` at standard rate; no separate thinking field in usage
- **Google**: Thinking tokens reported in dedicated `usageMetadata.thoughtsTokenCount` field; **critical inconsistency**: on Gemini Developer API, `candidatesTokenCount` includes thinking tokens, while on Vertex AI it does not

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Priority 5: API response field verification

OpenAI — confirmed with additions

The fields `usage.prompt_tokens`, `usage.completion_tokens`, and `usage.total_tokens` **remain accurate** for the Chat Completions API. Microsoft Learn New sub-fields have been added:

```
usage.completion_tokens_details.reasoning_tokens     // NEW
usage.completion_tokens_details.accepted_prediction_tokens  // NEW
usage.completion_tokens_details.rejected_prediction_tokens  // NEW
usage.prompt_tokens_details.cached_tokens           // NEW
```

Important: The newer **Responses API** uses different field names: `input_tokens`/`output_tokens` instead of `prompt_tokens`/`completion_tokens`, with corresponding `input_tokens_details`/`output_tokens_details`.

Anthropic — confirmed with additions

The fields `usage.input_tokens` and `usage.output_tokens` **remain accurate** for Claude 3.7 and all Claude 4 models. New fields:

```
usage.cache_creation_input_tokens          // NEW
usage.cache_read_input_tokens              // NEW
usage.server_tool_use.web_search_requests  // NEW
```

No separate `thinking_tokens` field exists—thinking tokens are included in `output_tokens`. Thinking content appears as content blocks with `type: "thinking"` in the response body.

Google Gemini — confirmed with significant additions

The fields `usageMetadata.promptTokenCount` and `usageMetadata.candidatesTokenCount` **remain accurate** for Gemini 2.x. Major new fields:

```
usageMetadata.thoughtsTokenCount           // NEW - thinking/reasoning tokens
usageMetadata.toolUsePromptTokenCount      // NEW - tool execution tokens
usageMetadata.promptTokensDetails[]        // NEW - per-modality breakdown
usageMetadata.candidatesTokensDetails[]    // NEW - per-modality output
```

⚠️ **Critical API inconsistency**: On the Gemini Developer API, `candidatesTokenCount` **includes** thinking tokens. On Vertex AI, it **excludes** thinking tokens (they appear only in `thoughtsTokenCount`). The whitepaper should document this discrepancy.

Consolidated model pricing reference table for WP-003

This table reflects verified pricing as of March 18, 2026 from official sources. Models marked with † are deprecated but may still function.

Provider	Model	Input/1M	Output/1M	Context	Notes
OpenAI	GPT-5.4	$2.50	$15.00	1.05M	Current flagship (Mar 2026)
OpenAI	GPT-4.1	$2.00	$8.00	1.05M	Best value full-size
OpenAI	GPT-4.1 nano	$0.10	$0.40	1.05M	Cost-optimized agents
OpenAI	GPT-4o	$2.50	$10.00	128K	Still available in API
OpenAI	GPT-4o mini	$0.15	$0.60	128K	Legacy cost leader
OpenAI	o3	$2.00	$8.00	200K	Reasoning (incl. thinking)
OpenAI	o4-mini	$1.10	$4.40	200K	Cost-effective reasoning
Anthropic	Claude Opus 4.6	$5.00	$25.00	1M	Current flagship (Feb 2026)
Anthropic	Claude Sonnet 4.6	$3.00	$15.00	1M	Best value Anthropic
Anthropic	Claude Haiku 4.5	$1.00	$5.00	200K	Fastest Anthropic
Anthropic	Claude 3.5 Haiku †	$0.80	$4.00	200K	Deprecated/retired
Google	Gemini 3.1 Pro Prev.	$2.00	$12.00	1M	Latest (preview)
Google	Gemini 2.5 Pro	$1.25	$10.00	1M	Stable flagship
Google	Gemini 2.5 Flash	$0.30	$2.50	1M	Best value w/ thinking
Google	Gemini 2.5 Flash-Lite	$0.10	$0.40	1M	Most cost-effective
Google	Gemini 2.0 Flash †	$0.10	$0.40	1M	Shutdown Jun 1, 2026
Groq	Llama 4 Scout	$0.11	$0.34	128K	Fastest inference
Groq	Llama 4 Maverick	$0.20	$0.60	128K	Best open MoE
Groq	Llama 3.3 70B	$0.59	$0.79	128K	Replaces 3.1 70B
Together	Llama 4 Maverick	$0.27	$0.85	128K	—
Together	Llama 3.3 70B	$0.88	$0.88	128K	—
Fireworks	Llama 70B-class	~$0.90	~$0.90	128K	Tiered pricing
DeepSeek	V3.2	$0.28	$0.42	128K	Chat + reasoning

Conclusion and recommended whitepaper actions

The whitepaper requires updates in three categories. First, **model references must be modernized**: replace Claude 3.5 Haiku with Claude Haiku 4.5, replace Llama 3.1 70B with Llama 3.3 70B (or Llama 4 Scout for lowest cost), and add GPT-4.1 and GPT-5.4 to the model table. Second, **the cost illustration table needs recalculation**—the original figures assumed pricing that no longer exists (particularly for Groq's Llama offering), and the spread between models has narrowed. Third, **API response field documentation should be expanded** to cover reasoning/thinking token fields (`completion_tokens_details.reasoning_tokens` for OpenAI, `thoughtsTokenCount` for Gemini) and the critical Gemini Developer API vs. Vertex AI inconsistency in thinking token counting.

The most surprising finding is the **convergence in pricing**: frontier models from all three providers now cluster around **$2–3 input / $10–15 output per 1M tokens** with 1M context windows, while cost-optimized tiers have dropped to **$0.10–0.30 input / $0.40–2.50 output**. The 100× cost spread between frontier and economy models creates significant optimization opportunities for ContextKeeper's token intelligence layer—arguably more than when the original whitepaper was drafted.